Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18CIC33]

# THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

## COMPUTER ORGANIZATION AND ARCHITECTURE

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

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| **1.** | **a.** | Illustrate how processor communicates with main memory with the necessary diagram. | **08** |
| | **b.** | Define Complex Instruction Set Computer (CISC) & Reduced Instruction Set Computer (RISC) and compare them. | **06** |
| **2.** | **a.** | Analyze basic performance equation and computer performance measurement | **08** |
| | **b.** | **i.** Using 16-bit address lines and 32-bit address lines how much memory can be accessed.<br>**ii.** Calculate the number of address lines required to access 64KB memory.<br>**iii.** Define Link register (LR) and stack pointer (SP). | **06** |
| **3.** | **a.** | Discuss carry save adder and multiply 45*63 using carry save adder. | **08** |
| | **b.** | With a block diagram illustrate the basic organization of a micro programmed control unit | **06** |
| | | **OR** | |
| **4.** | **a.** | Develop Booth's Multiplication algorithm, Solve (+13) * (-6) using same. | **08** |
| | **b.** | Demonstrate three bus organization of a data path with a neat diagram. | **06** |
| **5.** | **a.** | Compose different memory mapping techniques with neat diagram | **10** |
| | **b.** | Differentiate between DRAM and SRAM. | **04** |
| | | **OR** | |
| **6.** | **a.** | What is asynchronous DRAM? Model the internal organization of a 2M*8 dynamic memory chip. | **08** |
| | **b.** | Discuss memory hierarchy with neat diagram also specify the requirement for memory hierarchy. | **06** |
| **7.** | **a.** | Discuss different registers in keyboard & display interfaces and write a code to read a line from the keyboard & store it in buffer & echoes it back to the display. | **10** |
| | **b.** | List and discuss the registers used in DMA interface | **04** |
| | | **OR** | |
| **8.** | **a.** | What is bus arbitration? Discuss different bus arbitration techniques. | **08** |
| | **b.** | What are the objectives of Universal Serial Bus (USB), Discuss USB tree structure with diagram? | **06** |